

OMKAR WAGHOLIKAR

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EDUCATION

Pune Institute of Computer Technology July 2021 - July 2025
Bachelor of Engineering - Computer Engineering (Hons. Data Science and Visualization) CGPA: 8.8

TECHNICAL SKILLS

Languages: Java, Python, JS, C#, TypeScript
Databases: PostgreSQL, MongoDB, MySQL, Redis
Frameworks: Django, DRF, React, Node.js, Celery
Observability: OpenTelemetry, Distributed Tracing
Data Engineering: Airflow, DBT, BigQuery, GCS
Technologies: Docker, Kafka, RabbitMQ, Elastic Search
AI Tools: Claude Code, GitHub Copilot, AWS Bedrock

EXPERIENCE

Backend Software Development Engineer, Intangles June 2026 – Current

- Boosted throughput by **4x** for the org’s most-used event consumer by using **Redis-cached short-circuit** and a **heuristics** based approach with push invalidation to minimize redundant compute.
- Owned end-to-end **OpenTelemetry** rollout across all, establishing the entire org’s **observability** baseline.
- Drove a **platform stability** initiative on **TimescaleDB**: profiled consumption patterns, cut redundant reads and double-count queries, and shipped several database optimizations.

Associate Software Development Engineer, NiCE Ltd July 2025 – May 2026

- Optimized core backend application logic in **C#**, improving **performance and reliability** of critical paths.
- Fixed infinite reprocessing issue in **DLQ** by debugging and correcting **routing logic** to prevent retries.
- Designed document ingestion pipelines with checkpointing, batch processing, and retry/failure handling, using **AWS Bedrock, Lambda, and Batch** for scalable, event-driven solutions.

Project Assistant, Pune Knowledge Cluster October 2024 – February 2025

- **Architected and deployed a full-stack platform** for a **national-level** data challenge competition.
- Designed a **scalable submission pipeline** handling large user file uploads via **Azure object storage**.
- Created internal tools for communication, contact registration, and **workflow automation**.

Software Engineering Intern, Digital and Technology, General Mills India June 2024 – August 2024

- Worked with the **Data Engineering** team on a commercial **File Data Ingestion** project utilizing BigQuery.
- Established standardized procedures for incoming file management via shared mailbox and **GCP Buckets**.
- Executed the FDI process with **data quality checks** using **DBT**, integrated into **Airflow DAGs**.

Research Associate, Pune Knowledge Cluster November 2023 – February 2024

- Developed solutions for Central Government Problem Statement using **Docker, Elastic Search, and DRF**.
- Led the **full project lifecycle** from requirement analysis to system design, development, and deployment.
- Implemented a custom **Information Retrieval System** using **RAG** to assist social healthcare workers.

PROJECTS

CloudDrive: Self-Hosted Cloud Storage (Go, Kotlin/Jetpack Compose, SQLite, SeaweedFS, Kafka)

- Built a self-hosted, **Google Drive-style** storage system for low-RAM ARM devices: a **Go/Gin REST API** with a native Android client (**Jetpack Compose, Media3 ExoPlayer**).
- Implemented **resumable chunked uploads, Kafka-driven** thumbnailing, **token-based auth** with rotating refresh tokens, and **LAN-aware transfer**.

Project MAVIS: Media Authenticity Verification and Integrity System (Django, OpenCV, PyTorch)

- Designed a dual-channel verification scheme (EXIF metadata + embedded QR watermark) that degrades gracefully when metadata is stripped
- Established **provenance and media authenticity** via **RSA-signed** device-level image hashing, **hardware-backed security**, and **steganographic and cryptographic** techniques.
- Engineered a multi-layered approach using Wavelet-DCT, GAN-based steganographic watermarking, and Reed-Solomon encoding, plus **QR-embedded provenance** and tamper-proof certificate generation.

BRAGS: Build your own RAGs (Python, Go, LangChain)

- **Python package** for easily deploying custom **Retrieval-Augmented Generation (RAG)** pipelines.
- Combines RAG with a background **Go file watcher** tracking OS-level events for **real time** synchronization.
- Natively supports **Claude MCP**, exposing an **MCP server** so any RAG pipeline is directly queryable.

EXTRA-CURRICULAR ACTIVITIES

- **Open Source Contributor:** [CircuitVerse](#): **3 merged PRs** (bug fixes, refactoring) and **1 issue** reported. [AeroFTP](#): **3 features co-authored** (theming, security, UI layout)
- Accepted speaker for a talk on [Unix file descriptors and PTY internals](#) at [FOSS United Pune](#), April 2026.

CERTIFICATIONS

[Deep Learning Specialization](#) | [Basic GANs](#) | [GCP Core Infrastructure](#) | [LFD103: Linux Kernel Development](#)